A Classical Reasoning System for Entailment over Rules, Causal Graphs and Logic Stories

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1 Overview

The system decides whether a query is entailed by a natural-language knowledge base and outputs `True` or `False`. The pipeline is:

text $\rightarrow$ **router** $\rightarrow$ **translator** $\rightarrow$ (facts, rules, interventions, goal) $\rightarrow$ **forward chaining** $\rightarrow$ **True/False**

One reasoning engine serves all three public datasets (RuleTaker, CounterBench, FOLIO). Only the translator in front of the engine differs. The translators are regular-expression templates, so no generative model is used at any point, and the final decision always comes from the classical engine.

2 Representation

An *atom* is a hashable name for a proposition. A *literal* is a pair (atom, polarity). A *rule* is a pair (body, head): the head atom holds if every literal in the body holds. A knowledge base is a set of fact atoms plus a list of rules, and a query is a goal atom plus an optional set of *interventions* (atoms forced to True or False from outside). Each translator maps its dataset into this form:

Dataset	Atoms	How text becomes facts and rules
RuleTaker	normalised sentence, e.g. `river is fast`	Lines under “Facts:” become facts. Each rule line (five templates such as “Assuming A, we can say B” or “If A, then B”) becomes one rule $A \rightarrow B$. The statement becomes the goal.
CounterBench	variable name, e.g. `Lumbo`	“A causes B” gives $A \rightarrow B$; “A and B together cause C” gives one rule with two body literals; “A or B causes C” gives two rules; “A causes not B” gives a rule whose body is negated (De Morgan), so B holds when A is absent. Variables that are never an effect are root facts unless observed false. The question gives the target atom and the interventions (e.g. “not Ziklo” forces Ziklo to False; nested questions give two interventions).
FOLIO	(predicate, entity) pair	“X is a Y” sentences become facts; “All A are B” is grounded into one rule per named entity. Sentences containing or / not / some / either and similar words are skipped, so the translation stays sound but incomplete.

The router picks the translator from the text: “Facts:” and “Rules:” markers mean RuleTaker; the word “cause” in the context together with “occur” in the question means CounterBench; everything else is treated as FOLIO.

3 Reasoning algorithm

The engine is **forward chaining to a fixpoint**. Each pass recomputes the derived set from the facts, applies every rule whose body holds, and stops when nothing changes. A positive literal holds if its atom is derived; a negative literal holds if its atom is not derived (negation as failure, i.e. the closed-world assumption).

```
known = (facts - forced_false) + forced_true
repeat at most (len(rules) + 2) times:
    new = (facts - forced_false) + forced_true
    for (body, head) in rules:
        if head is not forced and every literal in body holds in known:
            add head to new
    if new == known: stop
    known = new
answer = (goal in known)
```

Recomputing from the facts in each pass keeps negation correct on the acyclic CounterBench graphs, because a negative literal is re-evaluated once its atom has settled. For the positive-only RuleTaker and FOLIO rules the same loop computes the ordinary least fixpoint. A *counterfactual* question is answered by running the same loop with the intervened atoms forced, which ignores their own causal rules (graph surgery). Each derived atom records the rule body that first produced it, which gives the reasoning traces in Section 5. The cost is at most passes $\times$ rules $\times$ body size, which is tiny for knowledge bases of 5 to 20 rules.

4 Experiments on the public data

All numbers use the public training splits only (RuleTaker 1000 rows, CounterBench 1000, FOLIO 1001); the private test set was never used. Accuracy counts a query as correct when the predicted flag equals the label; for FOLIO the label "Uncertain" is treated as not entailed. Times are the total reasoning time for the whole split.

Dataset	Accuracy	Breakdown	Time
RuleTaker	98.3% (983/1000)	depth 1: 100.0% (196/196); depth 2: 96.8% (393/406); depth 3: 99.0% (394/398)	0.09 s
CounterBench	96.0% (960/1000)	basic 100.0%; conditional 92.0%; joint 96.0%; nested 96.0% (250 each)	0.11 s
FOLIO	61.3% (614/1001)	same as always answering False (613/1001); see Section 6	0.09 s

Experiment 1: reading of nested counterfactuals. For questions such as "Assume not Ziklo, and based on this assumption, further suppose not Sline", I compared three readings on the 250 nested questions: apply both interventions (96.0%, 240/250), only the first (76.0%, 190/250), only the last (80.0%, 200/250). Applying both was kept.

Experiment 2: negated causes. The default reading treats "not Y" in a rule body as true whenever Y is not derived. I tested a stricter variant in which absence counts only if Y was forced false or is a root. It scored 70.5% (705/1000) on CounterBench (basic dropped from 100% to 62%), against 96.0% for the default, so it was rejected.

Efficiency. The whole public set (3001 queries) is answered in about 0.3 s of reasoning time.

5 Reasoning traces

Traces are produced by the engine itself (derived atom, then the rule body that produced it, then its supports), shown here in a shortened format.

RuleTaker, depth 3. On the knowledge base of a training example (facts include "The globe is friendly"), I posed the query "The globe writes stories." to show a three-step derivation; the engine derives it (True).

```
globe writes stories  <-  [mike runs marathons]
  mike runs marathons  <-  [dog runs marathons]
    dog runs marathons  <-  [globe is friendly]
      globe is friendly  [given]
```

RuleTaker, not derivable. Query: "The sandwich dances gracefully." The only rule mentioning it has it as a premise, and no rule concludes it, so it is never derived (False, matches the label).

CounterBench, conditional. "Ziklo causes not Blaf, Blaf and Trune together cause Vork, Vork causes Lumbo. We observed Trune." Query: would Lumbo occur if not Ziklo instead of Ziklo? Intervention: Ziklo = False (answer yes).

```
Lumbo  <-  [Vork]
  Vork  <-  [Blaf, Trune]
    Blaf  <-  [not Ziklo]      (Ziklo is forced False, so "not Ziklo" holds)
    Trune  [given]             (observed root)
```

CounterBench, basic. “Ziklo causes Blaf, Blaf causes Trune, Trune causes Vork, Vork causes Lumbo.” With Ziklo forced False nothing in the chain is derived, so Lumbo does not occur (answer no).

6 Error analysis and limitations

RuleTaker (17 errors). I inspected two of them. In both, the statement follows from one stated rule and a stated fact, for example “Assuming the train plays instruments, we can say the artist is fast” with the fact “The train plays instruments”, yet the label is False. These look like label noise or a semantic the data uses that the engine does not model. I also skimmed the other 15 and they appear to be label noise as well, but I did not check each one in detail.

CounterBench (40 errors, 20 of them conditional). The three conditional errors I inspected share a pattern: a negated cause (“X or not Y causes Z”) together with an observation on a variable that is itself an effect. The translator currently ignores observations on non-root variables. Handling them properly (as evidence in the abduction step of counterfactual reasoning) is the first thing I would try with more time; this explanation is a hypothesis from three examples and is not tested.

FOLIO (61.3%). FOLIO is free-form first-order logic in English. The translator handles only simple “X is a Y” facts and “All A are B” rules, and in practice it parsed 230 of 1001 conclusions and derived True for exactly one, so the result equals the constant-False baseline. The engine is not the bottleneck; the English-to-logic translation is. A realistic extension is a proper first-order parser with resolution or a tableau prover. Treating “Uncertain” as not entailed is an assumption.

Generality. The engine is general (any rule set over atoms with negation and interventions), but the three translators are templates written for the formats seen in the public data. A test set in a different phrasing would need new templates, and any sentence a translator cannot read is skipped, which can only lower accuracy.